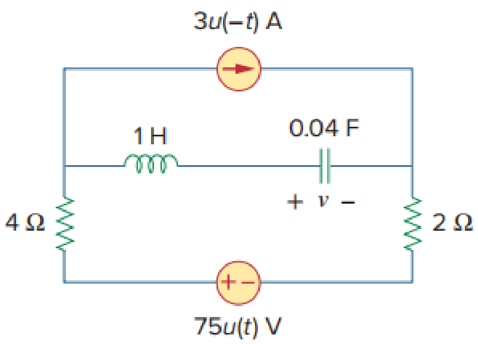
Chapter 8, Problem 32P

Problem

For the circuit in Fig. 8.80, find $v(t)$ for $t > 0$.

Figure 8.80:



Step-by-step solution

Step 1 of 7

Refer to the Figure 8.80 in the textbook.

For $t < 0$,

$u(t) = 0$ and $u(-t) = 1$

For $t > 0$,

$u(t) = 1$ and $u(-t) = 0$

Therefore for $t < 0$,

$3u(-t) = 3\text{A}$ and $75u(t) = 0\text{V}$

For $t < 0$ the current source of 3 A is on while the voltage source $75u(t)$ is short circuited and the circuit is under dc steady state.

At steady state the inductor behaves as short circuit and the capacitor behaves as open circuit.

Therefore, the modified circuit diagram at $t < 0$ is shown in Figure 1.

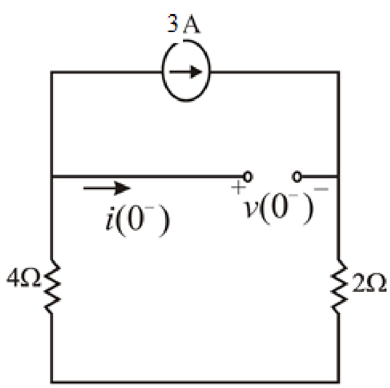


Figure 1

Step 2 of 7

From the Figure 1,

The current in the inductor is,

$$i(0^-) = 0\,\text{A}$$

Calculate the voltage across the capacitor.

$$v(0^-) + 3(2) + 4(3)$$
$$v(0^-) = -18\,\text{V}$$

Due to the continuity of the capacitor voltage and inductor current

$$i(0^+) = i(0^-)$$

$$i(0^+) = 0\text{A}$$

$$v(0^+) = v(0^-)$$

$$v(0^+) = -18\ \text{V}$$

Step 3 of 7

For $t > 0,\ 3u(-t) = 0\text{A}$ and $75u(t) = 75\ \text{V}$.

At $t > 0,$ the 3 A current source is open circuited and the voltage source $75u(t)$ is on.

Therefore, the reduced circuit becomes a series LCR circuit as shown in Figure 2.

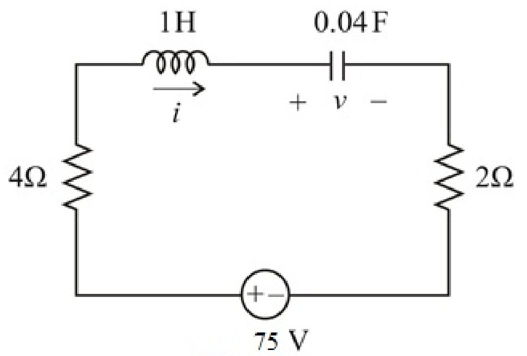


Figure 2

Step 4 of 7

The circuit in Figure 2 is a series *RLC* circuit.

For a series *RLC* circuit,

$$\alpha = \frac{R}{2L}, \quad \omega_o = \frac{1}{\sqrt{LC}}.$$

Calculate the value of α.

$$\alpha = \frac{R}{2L}$$

Substitute 6Ω for *R*, 1 H for *L*.

$$\alpha = \frac{6}{2} = 3$$

Calculate the value of ω_o.

$$\omega_o = \frac{1}{\sqrt{LC}},$$

Substitute 1 H for *L*, 0.04 F for *C*.

$$\omega_0 = \frac{1}{\sqrt{0.04}} = 5$$

Step 5 of 7

Here $\alpha < \omega_0$, therefore this is an underdamped case.

For the underdamped case the voltage $v(t)$ across the capacitor is given as:

$$v(t) = V_S + (A_1 \cos \omega_d t + A_2 \sin \omega_d t)\, e^{-\alpha t} \quad \ldots\ldots (1)$$

Here,

V_S is the final value of the voltage across the capacitor.

At steady state, the capacitor is replaced by an open circuit a shown in Figure 3.

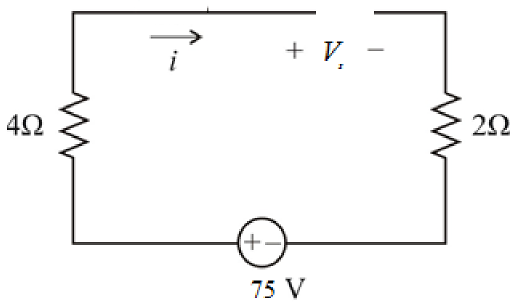


Figure 3

Calculate the steady state voltage V_S.

$$V_s = 75\,\text{V}$$

Step 6 of 7

Calculate the value of ω_d.

$$\omega_d = \sqrt{\left|\alpha^2 - \omega_0^2\right|}$$
$$= \sqrt{\left|3^2 - 5^2\right|}$$
$$= 4$$

Substitute 4 for ω_d, 3 for α and 75 V for V_S in equation (1).

$$v(t) = 75 + \left(A_1 \cos 4t + A_2 \sin 4t\right) e^{-3t} \text{ ...... (2)}$$

Use the initial values $v(0^-) = 18\,\text{V}$ and $i_L(0^-) = 0\,\text{A}$ to calculate the value of constants A_1 and A_2.

Substitute 0 for *t*, $-18\,\text{V}$ for $v(0^-)$ in equation (2).

$$v(0) = 75 + \left(A_1 \cos 0 + A_2 \sin 0\right)$$
$$-18 = 75 + A_1$$
$$A_1 = -93$$

Step 7 of 7

The current through capacitor at $t = 0$ is,

$$i_L(0) = C\frac{dv(0)}{dt}$$
$$0 = (0.04)\frac{dv(0)}{dt}$$
$$\frac{dv(0)}{dt} = 0\,\text{V/s}$$

Differentiate the equation (2).

$$\frac{dv}{dt} = \left(-4A_1 \sin 4t + 4A_2 \cos 4t\right)e^{-3t} - 3e^{-3t}\left(A_1 \cos 4t + A_2 \sin 4t\right)$$

Substitute 0 for *t*, -93 for A_1.

$$\frac{dv(0)}{dt} = 4A_2 - 3(-93)$$
$$0 = 4A_2 + 279$$
$$A_2 = -69.75$$

Substitute -93 for A_1 and -69.75 for A_2 in equation (2).

$$v(t) = 75 + \left(-93 \cos 4t - 69.75 \sin 4t\right) e^{-3t}\ \text{V}$$

Therefore, the voltage $v(t)$ across the capacitor is $\boxed{75 + \left(-93 \cos 4t - 69.75 \sin 4t\right) e^{-3t}\ \text{V}}$.